Customers like you

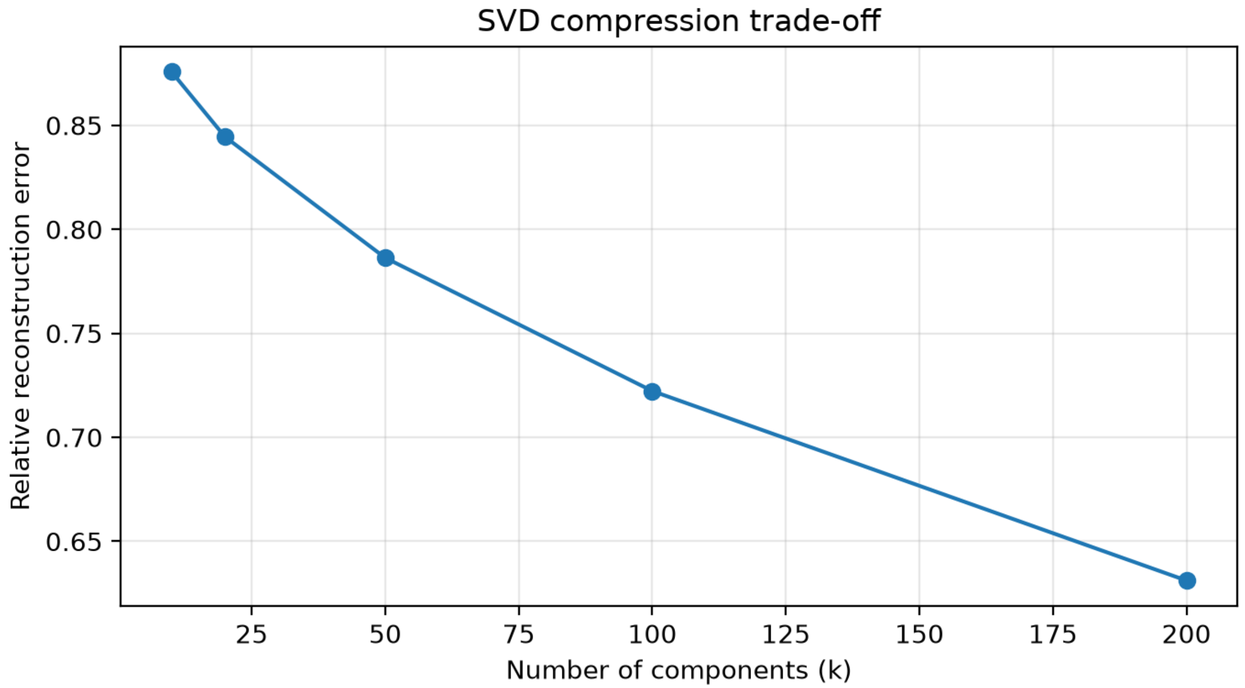
Retail Recommendations and Customer Segmentation
DS-5020 Mini-Project 01
Prathmesh Deepak Pise | Individual submission

Data & Preparation

- • Online Retail II: 1,067,371 transaction lines
- • Removed cancellations, invalid quantities/prices, missing customer IDs, and non-product entries
- • Retained 802,652 purchase lines
- • Reserved the final two months for testing
- • Training matrix: 5,482 customers × 3,690 products
- • Matrix is 97.95% sparse

SVD: Method & Compression

- • SVD works directly on sparse purchases without centering
- • Tested 10, 20, 50, 100, and 200 components
- • Selected 50 as a storage-focused compromise
- • Storage: 161.83 MB dense; 5.00 MB sparse; 3.67 MB compressed factors
- • Relative reconstruction error: 0.7863

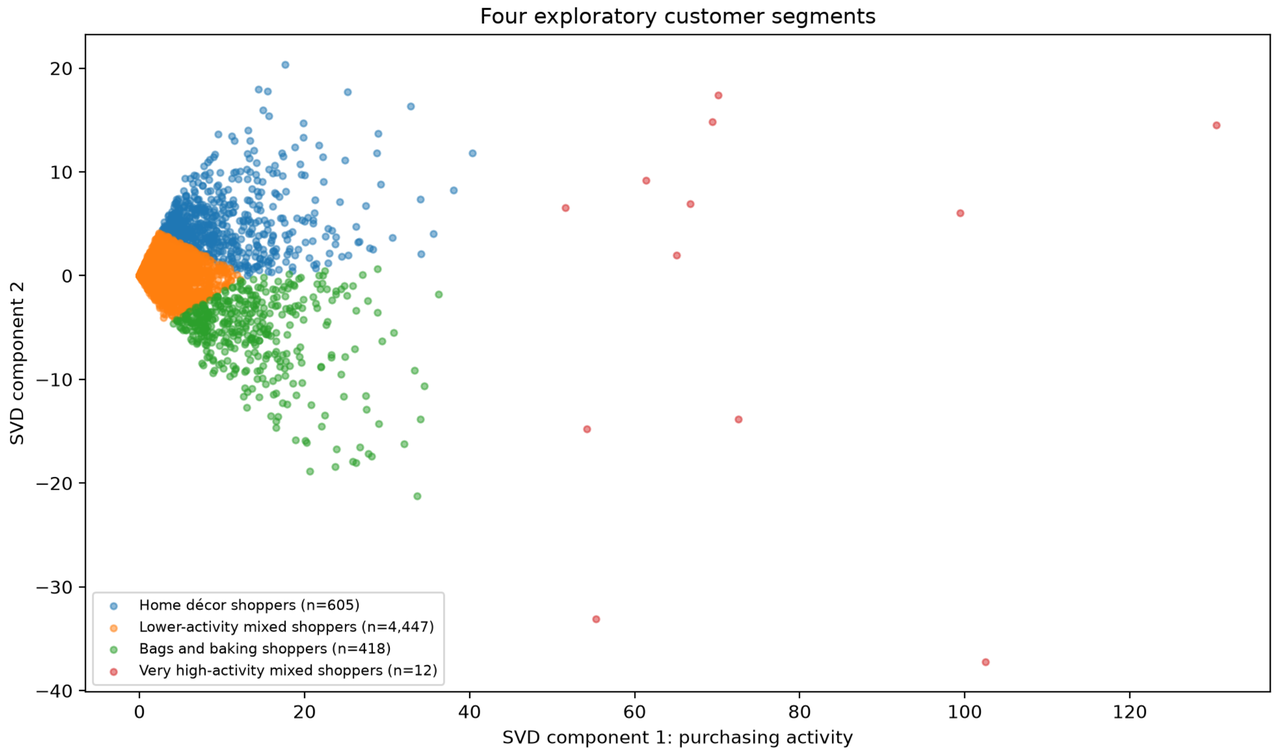


Recommendation Results

- • Hit rate at 10:
  SVD: 22.30% | PCA: 22.25% | Best sellers: 17.38%
- • Evaluated on the same 1,910 customers
- • Previously purchased products excluded
- • SVD improved on best sellers by 4.92 percentage points
- • PCA and SVD differed by only one successful customer
- • Recommendation generation took 2.29 second
  Excludes data preparation, fitting, and saving

Customer Groups

- • Home decor shoppers: 605
- • Lower-activity mixed shoppers: 4,447
- • Bags and baking shoppers: 418
- • Very high-activity mixed shoppers: 12
  Exploratory groups—purchasing activity strongly influences the map.



Limitations & Next Steps

- • Evaluation covers 71.45% of new test purchase pairs
- • New customers and excluded products need fallback recommendations
- • Compression loses substantial purchase detail
- • Customer groups overlap; the smallest contains only 12 customers
- • Offline hit rate does not prove increased sales
- • Next: stock checks, more test periods, marketing review, and a controlled customer trial